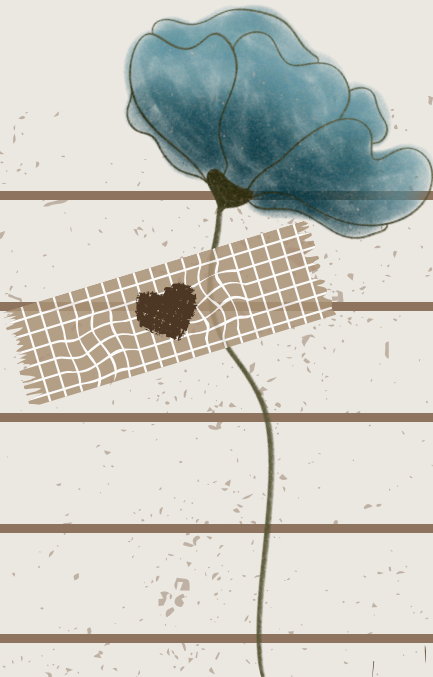




DIGITAL SIGNAL PROCESSING PRELIMS

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SIGNALS

- Convey information from one point to another (They may be generated by electronic means, or by some natural means such as talking, walking, your heart beating, an earthquake, the sun heating the sidewalk)
- described by a function of one or more independent variables. The value of the function (i.e., the dependent variable) can be real-valued scalar quantity a complex-valued quantity or perhaps a vector.



Examples:

$$s_1(t) = A \sin 3\pi t$$

$$s_2(t) = 5t$$

$$s_3(t) = 5t^2$$

$$s_4(x, y) = 3x + 2xy + 10y^2$$

$$s_5(t) = Ae^{j3\pi t} = A \cos 3\pi t + jA \sin 3\pi t$$

$$S_6(t) = \begin{bmatrix} s_1(t) \\ s_2(t) \\ s_3(t) \end{bmatrix}$$

SYSTEM

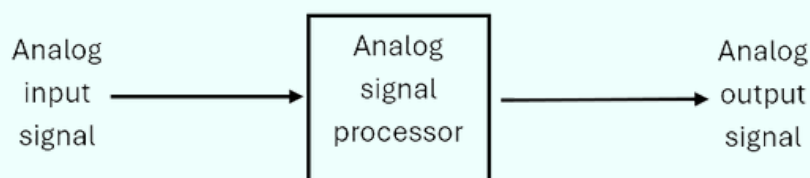
- process signals to produce a modified or transformed version of the original signal
- defined as a physical device that performs an operation on a signal.

Example:

A filter used to reduce the noise and interference corrupting a desired information-bearing signal is called a system.

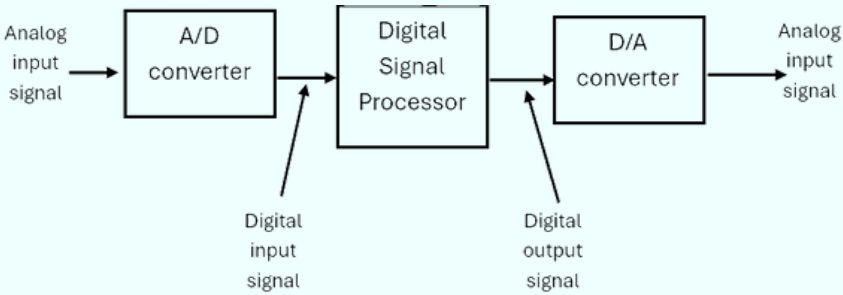
BASIC ELEMENTS OF A DIGITAL SIGNAL PROCESSING SYSTEM

Most of the signals encountered in science and engineering are analog in nature. That is, the signals are functions of a continuous variable, such as time or space, and usually take on values in a continuous range. Such signals may be processed directly by appropriate analog systems (such as filters, frequency analyzers, or frequency multipliers) for the purpose of changing their characteristics or extracting some desired information.



In such a case we say that the signal has been processed directly in its analog form, as illustrated in figure above. Both the input signal and the output signal are in analog form.





- There are many reasons why digital signal processing of an analog signal may be preferable to processing the signal directly in the analog domain, as mentioned briefly earlier. First, a digital programmable system allows flexibility in reconfiguring the digital signal processing operations simply by changing the program.
- Digital signals are easily stored on magnetic media (tape or disk) without de-terioration or loss of signal fidelity beyond that introduced in the A/D conversion.
- One practical limitation is the speed of operation of A/D converters and digital signal processors. We shall see that signals having extremely wide bandwidths require fast-sampling-rate A/D converters and fast digital signal processors. Hence there are analog signals with large bandwidths for which a digital processing approach is beyond the state of the art of digital hardware.

MULTICHANNEL AND MULTIDIMENSIONAL SIGNALS

As explained in Section 1, a signal is described by a function of one or more independent variables. The value of the function (i.e., the dependent variable) can be a real-valued scalar quantity, a complex-valued quantity, or perhaps a vector. For example, the signal:

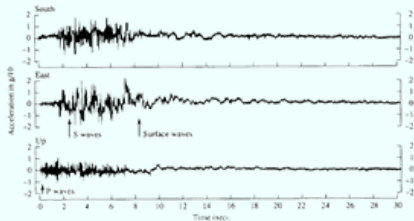
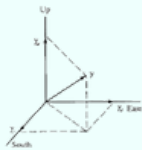
$$s_1(t) = A \sin 3\pi t$$

is a real-valued signal. However, the signal

$$s_2(t) = Ae^{j3\pi t} = A \cos 3\pi t + j A \sin 3\pi t$$

is complex valued.

In some applications, signals are generated by multiple sources of multiple sensors. Such signals, in turn, can be represented in vector form. Figure below shows the three components of a vector signal that represents the ground acceleration due to an earthquake.



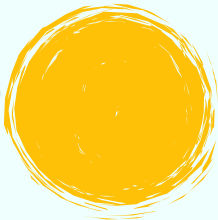
This acceleration is the result of three basic types of elastic waves. The primary (P) waves and the secondary (S) waves propagate within the body of rock are longitudinal and transversal, respectively. The third type of elastic wave is called the surface wave, because it propagates near the ground surface. If $S_k(t)$, $k = 1, 2, 3$, denotes the electrical signal from the k th sensor as a function of time, the set of $p = 3$ signals can be represented by a vector $S_3(t)$, where

$$S_3(t) = \begin{bmatrix} S_1(t) \\ S_2(t) \\ S_3(t) \end{bmatrix}$$

We refer to such a vector of signals as a multichannel signal. In electrocardiography, for example, 3-lead and 12-lead electrocardiograms (ECG) are often used in practice, which result in 3-channel and 12-channel signals.



CONTINUOUS-TIME VERSUS DISCRETE-TIME SIGNALS



Signals can be further classified into four different categories depending on the characteristics of the time (independent) variable and the values they take.



Continuous Time Signals

Or analog signals are defined for every value of time and they take on values in the continuous interval (a, b), where
a can be $-\infty$ &
b can be ∞ .

Mathematically, these signals can be described by functions of a continuous variable.

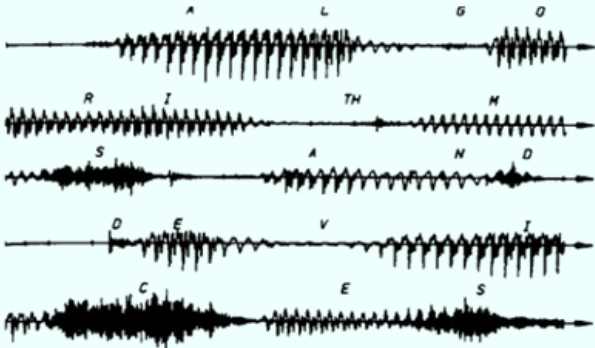


Figure 1.1
Example of a speech signal.

Discrete-time signals

Are defined only at certain specific values of time. These time instants need not be equidistant, but in practice they are usually taken at equally spaced intervals for computational convenience and mathematical tractability.

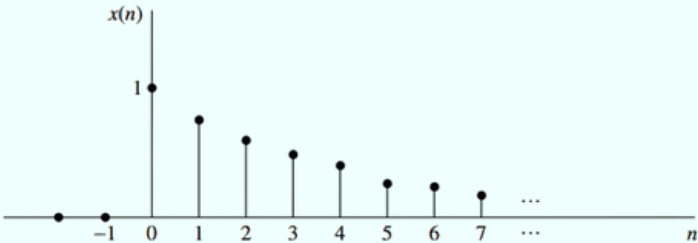
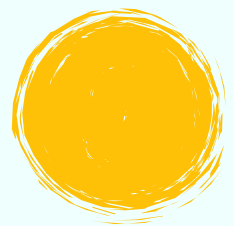


Figure 2.3 Graphical representation of the discrete time signal $x(n) = 0.8^n$ for $n > 0$ and $x(n) = 0$ for $n < 0$.



CONTINUOUS-VALUED VERSUS DISCRETE- VALUED SIGNALS



The values of a continuous-time or discrete-time signal can be continuous or discrete

Continuous-Valued signal

If a signal takes on all possible values on a finite or an infinite range.

Discrete-Valued signal

If the signal takes on values from a finite set of possible values.



DETERMINISTIC VERSUS RANDOM SIGNALS



The mathematical analysis and processing of signals requires the availability of a mathematical description for the signal itself. This mathematical description, often referred to as the signal model, leads to another important classification of signals.

Deterministic

Any signal that can be uniquely described by an explicit mathematical expression, a table of data, or a well-defined rule is called deterministic. This term is used to emphasize the fact that all past, present, and future values of signals are known precisely, without any uncertainty.

Random

The lack of such a relationship implies that such signals evolve in time in an unpredictable manner.

We refer to these signals as random. The mathematical framework for the theoretical analysis of random signals is provided by the theory of probability and stochastic process.

DISCRETE TIME SIGNALS

A discrete-time signal $x(n)$ is a function of an independent variable that is an integer. It is graphically represented in the figure before. It is important to note that a discrete-time signal is not defined at instants between two successive samples. Simply, the signal $x(n)$ is not defined for noninteger values of n .

Besides the graphical representation of a discrete-time signal or sequence, there are some alternative representations that are often more convenient to use. These are:

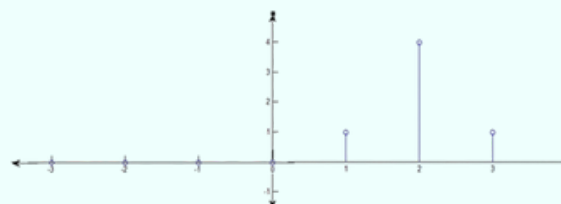


Functional Representation

$$x(n) = \begin{cases} 1, & \text{for } n = 1, 3 \\ 4, & \text{for } n = 2 \\ 0, & \text{elsewhere} \end{cases}$$

Graphical Representation

Graphically represented as



Okay, let's imagine we have a sequence of numbers, and we want to represent them in a certain way. In this case, we're looking at a graph where each position in the graph represents a specific value for a number, like a list of points.

Now, in this special kind of representation, we start by putting the number '1' in certain spots on the graph. Specifically, we put '1' at positions where $n=1$, $n=2$, and $n=3$. So, the first three spots in the graph will all have the value '1'.

Next, at position $n=2$, we're going to put a '4' instead of '1'. This is like saying, "Hey, at this specific spot on the graph, the number is special and should be 4 instead of 1."

Finally, for all the other positions on the graph that aren't mentioned, we'll just leave them as '0'. So, if there's a spot for $n=4$, $n=5$, and so on, those values will be '0', which basically means "nothing" or "no signal" in this case.

Tabular Representation

n	...	-2	-1	0	1	2	3	4	5
$x(n)$	...	0	0	0	1	4	1	0	0

Sequence Representation

An infinite-duration signal or sequence with the time origin ($n = 0$) indicated by the symbol $\uparrow$ is represented as

$$x(n) = \{\dots 0, 0, 0, 1, 4, 1, 0, 0, \dots\}$$

$\uparrow$

A sequence $x(n)$, which is zero for $n < 0$, can be represented as

$$x(n) = \{0, 1, 4, 1, 0, 0, \dots\}$$

$\uparrow$

The time origin for a sequence $x(n)$, which is zero for $n < 0$, is understood to be the first (leftmost) point of the sequence.

A finite-duration sequence can be represented as

$$x(n) = \{3, -1, -2, 5, 0, 4, -1\}$$

$\uparrow$

whereas a finite-duration sequence that satisfies the condition $x(n) = 0$ for $n < 0$ can be represented as

$$x(n) = \{0, 1, 4, 1\}$$

$\uparrow$

The signal consists of seven samples or points (in time), so it's called or identified as a seven-point sequence. Similarly, the sequence given by the figure above is a four point sequence.



PERIODIC AND APERIODIC SIGNALS

A signal $x[n]$ is periodic with period N ($N > 0$) if and only if $x(n + N) = x(n)$ for all n .

If there is no value of N that satisfies the equation above, the signal is called non periodic or aperiodic.

Periodic signals are power signal.

N = fundamental period in discrete time. N must be a positive integer.

ω = fundamental angle of frequency or discrete angle of frequency. It is denoted as $\omega = 2\pi f$

Frequency is denoted as $f = \frac{K}{N}$

Transformation of the independent variable (time)

A signal $x(n)$ may be shifted in time by replacing the independent variable n by $n - k$, where k is an integer. If k is a positive integer, the time shift results in a delay of the signal by k units of time. If k is a negative integer, the time shift results in an advance of the signal by $|k|$ units in time.

A signal $x(n)$ is graphically illustrated in Fig. 1.9(a). Show a graphical representation of the signals $x(n - 3)$ and $x(n + 2)$.

Solution. The signal $x(n - 3)$ is obtained by delaying $x(n)$ by three units in time. The result is illustrated in Fig. 1.9(b). On the other hand, the signal $x(n + 2)$ is obtained by advancing $x(n)$ by two units in time. The result is illustrated in Fig. 1.9(c). Note that delay corresponds to shifting a signal to the right, whereas advance implies shifting the signal to the left on the time axis.

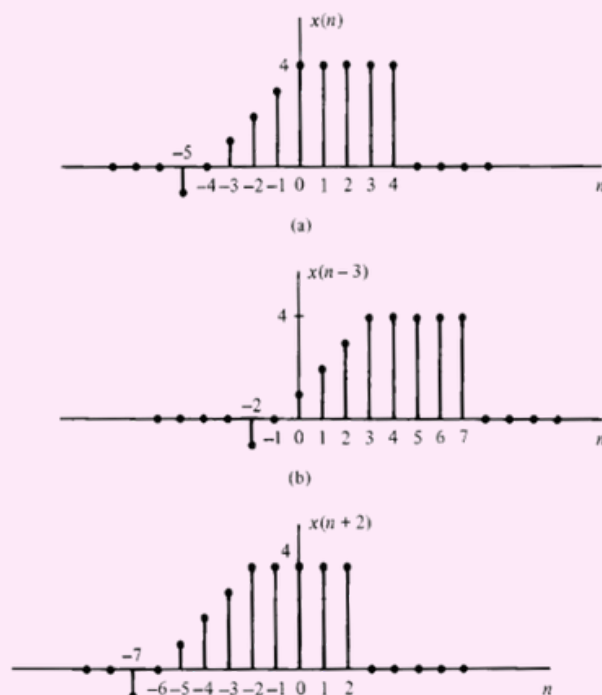


Figure 1.9
Graphical representation of a signal, and its delayed and

Folding Another useful modification of the time base is to replace the independent variable n by $-n$. The result of this operation is a folding or a reflection of the signal about the time origin $n = 0$

Time Shifting

A third modification of the independent variable which involves by replacing by μn , where μ is an integer.

If value μ is greater than 1, the signal is compressed.

If μ is a fraction, the signal is stretched.

Input/Output The input-output description of a discrete-time system consists of a mathematical expression or a rule, which explicitly defines the relation between the input and output signals (input-output relationship). The exact internal structure of the system is either unknown or ignored. Thus the only way to interact with the system is by using its input and output terminals (i.e., the system is assumed to be a “black box” to the user).

Aside from the previous figure and equation, alternatively, we use the notation $x[n] \xrightarrow{\mathcal{T}} y[n]$ which simply means that $y[n]$ is the response of the system $\mathcal{T}$ to the excitation $x[n]$. The following examples illustrate several different systems.

Determine the response of the following the systems to the input signal

$$x[n] = \begin{cases} |n| & , \quad -3 \leq n \leq 3 \\ 0 & , \quad \text{otherwise} \end{cases}$$

- a. $y[n] = x[n]$ (identity system)
- b. $y[n] = x[n - 1]$ (unit delay system)
- c. $y[n] = x[n + 1]$ (unit advance system)
- d. $y[n] = \frac{1}{3}[x[n + 1] + x[n] + x[n - 1]]$ (moving average filter)



Block Diagram Representation of Discrete-Time Systems

It is useful at this point to introduce a block diagram representation of discrete-time systems. For this purpose we need to define some basic building blocks that can be interconnected to form complex systems.

Adder – a system (adder) that performs the addition of two signal sequence to form another (the sum) sequence, which we denotes as $y[n]$. Note that it’s not necessary to store either one of the sequences in order to perform the addition. In other words, the addition operation is memoryless.

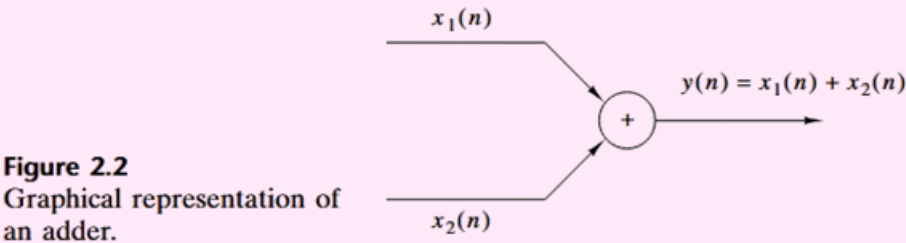


Figure 2.2
Graphical representation of an adder.

Constant multiplier – Simply represents applying a scale factor on the input $x[n]$. Note that this operation is also memoryless.

Signal multiplier – the multiplication of two sequences to form another (the product) sequence , denoted in the figure $y[n]$. As in the preceding two cases, we can view themultiplication operation as memoryless.

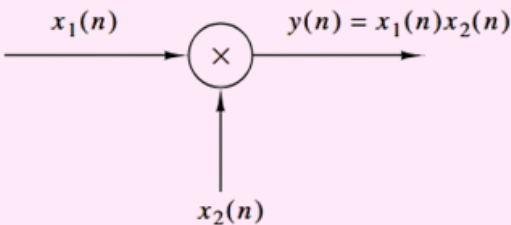
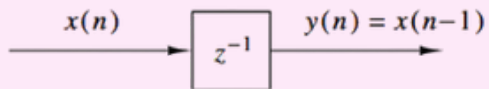


Figure 2.4
Graphical representation of a signal multiplier.

Unit Delay – The unit delay is a special system that simply delays the signal passing through it by one sample. If the input signal is $x[n]$, the output is $x[n-1]$. In fact, the sample $x[n-1]$ is stored in memory at a time $n-1$ and it is recalled from memory at time n to form:

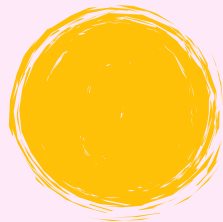
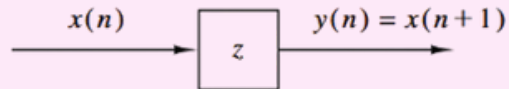
Thus this basic building block requires memory. The use of the symbol z^{-1} to denote the unit of delay will become apparent when discussing the z -transform.

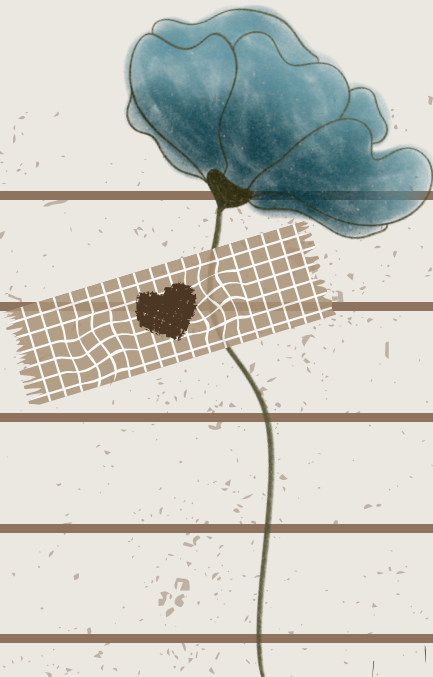
Figure 2.5
Graphical representation of
the unit delay element.



Unit advance element – In contrast to the unit delay, a unit advance moves the input $x[n]$ ahead by 1 sample in time to yield $x[n+1]$. We observe that any such advance is physically impossible in real time, since, in fact, it involves looking into the future of the signal. On the other hand, if we store the signal in the memory of the computer, we can recall any sample at any time. In such a non-real-time application, it is possible to advance the signal $x[n]$ in time.

Figure 2.6
Graphical representation of
the unit advance element.





DiGiTAL **SiGnAL** **P RocEssiNG** **MidT ERMs**

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CLASSIFICATION OF DISCRETE TIME SYSTEM

Static vs Dynamic



Static:

- A discrete-time system is called **static** or **memoryless** if its output at any instant n depends at most on the input sample at the same time, but not on past or future samples of the input.
- There is no need to store any of the past inputs or outputs in order to compute the present output.
- The system described by the following input-output equations

$$\begin{aligned} y[n] &= ax[n] \\ y[n] &= nx[n] + bx^3[n] \end{aligned}$$

are both static or memoryless.

Dynamic System:

- System that have a memory.
- If the output of the system at time n is completely determined by the input samples in the interval from $n-N$ to n ($N \geq 0$), the system is said to have memory of duration N .
- If $0 < N < \infty$, the system is said to have finite memory
- If $N = \infty$, the system is said to have a infinite memory.
- The system described by the following input-output equations

$$\begin{aligned} y[n] &= x[n] + 3x[n-1] & y[n] &= \sum_{k=0}^n x[n-k] \\ y[n] &= \sum_{k=0}^{\infty} x[n-k] \end{aligned}$$

are dynamic systems or system with memory.

time invariant vs time variant



Time Invariant:

- If its input-output characteristic do not change with time.
- **Definition.** A relaxed system is time-invariant or shift invariant if and only if

$$x[n] \xrightarrow{\mathcal{T}} y[n]$$

- implies that

$$x[n-k] \xrightarrow{\mathcal{T}} y[n-k]$$

- for every input signal $x[n]$ and every time shift k .

How to determine if the system is time variant or time invariant ? To determine if any given system is time invariant, we need to perform the test specified by the preceding definition.

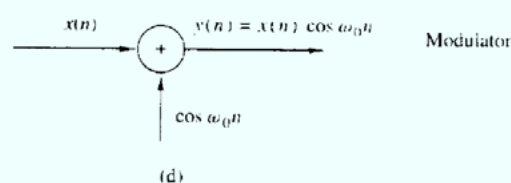
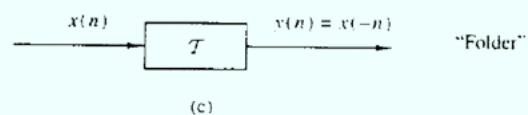
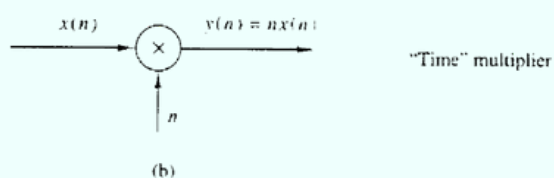
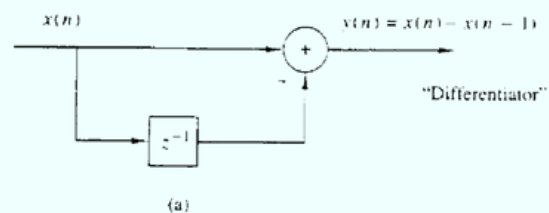
1. We excite the system with an arbitrary input sequence $x[n]$, which produces an output denoted as $y[n]$.

2. We delay the input sequence by some amount k and recomputed the output. In general, we can write the output as $y[n,k] = \mathcal{T}[x[n-k]]$.

3. If the output $y[n,k] = y[n-k]$, for all possible values of k , the system is time- invariant.

If the output $y[n,k] \neq y[n-k]$, even for one value of k , the system is time variant.

Example: Determine if the system shown below are time invariant or time variant.



CLASSIFICATION OF DISCRETE TIME SYSTEM

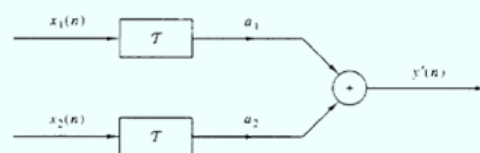
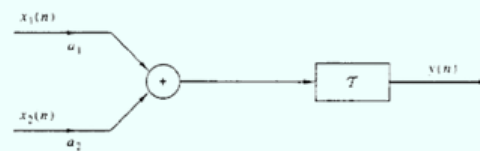


Linear:

- It is one that satisfies the *superposition principle*.
- A relaxed ,linear system with zero input produces a zero output
- **Superposition Principle** –requires that the response of the system to a weighted sum of signals be equal to the corresponding weighted sum of the responses (outputs) of the system to each of the individual input signals.
- **Definition.** A relaxed T system is linear if and only if
$$\mathcal{T}[a_1x_1[n] + a_2x_2[n]] = a_1\mathcal{T}[x_1[n]] + a_2\mathcal{T}[x_2[n]]$$
- for any arbitrary input sequence $x_1[n]$ and $x_2[n]$, and any arbitrary constant a_1 and a_2

Non-linear system:

- If a relaxed system does not satisfy the superposition principle as given by the definition above
- If a system produces a nonzero output with a zero input, the system may be either nonrelaxed or nonlinear
- T is linear if and only if $y[n]=y'[n]$



Casual vs. Non-casual

- **Definition.** A system is said to be causal if the output of the system at any time n depends only on present and past inputs, but does not depend on future inputs.
- If a system does not satisfy this definition, is called **noncausal**

- Stability is an important property that must be considered in any practical application of a system . Unstable systems usually exhibit erratic and extreme behavior and cause overflow in any practical implementation
- Definition. An arbitrary relaxed system is said to be bounded input-bounded output (BIBO) stable if and only if every bounded input produces a bounded output.



DISCRETE FOURIER TRANSFORM



- mathematical algorithm that can be computed efficiently on digital computers and other DSP Boards.
- an extension of CTFT for the time limited sequences with an additional restriction that the frequency is discretized at a finite set of values.

DFT Analysis Equation

$$x[k] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi \frac{kn}{N}}$$

DFT Synthesis Equation: (inverse DFT)

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} x[k] e^{j2\pi \frac{kn}{N}}$$

Example:



Calculate the four-point DFT of aperiodic sequence $x[k]$ of length $N = 4$ which is defined by:



Method 1



$$x[n] = \{2, 3, -1, 1\}$$

$$x[k] = \sum_{n=0}^3 x[n] e^{-j2\pi \frac{kn}{4}}$$

$$x[0] e^{-j2\pi \frac{(0)(0)}{4}} + x[1] e^{-j2\pi \frac{(0)(1)}{4}} + x[2] e^{-j2\pi \frac{(0)(2)}{4}} + x[3] e^{-j2\pi \frac{(0)(3)}{4}}$$

$$x[0] + x[1] + x[2] + x[3]$$

$$2 + 3 + (-1) + 1$$

$$5$$

DISCRETE FOURIER TRANSFORM (CONT.)

$$x[1] = \sum_{n=0}^3 x[n] e^{-j2\pi \frac{(1)(n)}{4}}$$

$$x[0] e^{-j2\pi \frac{(1)(0)}{4}} + x[1] e^{-j2\pi \frac{(1)(1)}{4}} + x[2] e^{-j2\pi \frac{(1)(2)}{4}} + x[3] e^{-j2\pi \frac{(1)(3)}{4}}$$

$$x[0] + x[1] e^{-j\frac{\pi}{2}} + x[2] e^{-j\pi} + x[3] e^{-j\frac{3\pi}{2}}$$

$$2 + (3) \left(\cos \frac{\pi}{2} - j \sin \frac{\pi}{2} \right) + (-1)(\cos \pi - j \sin \pi) + (1) \left(\cos \frac{3\pi}{2} - j \sin \frac{3\pi}{2} \right)$$

$$2 + 3(-j) + (-1)(-1) + (1)(j)$$

$$2 - 3j + 1 + j$$

$$x[2] = \sum_{n=0}^3 x[n] e^{-j2\pi \frac{(2)(n)}{4}}$$

$$x[0] e^{-j2\pi \frac{(2)(0)}{4}} + x[1] e^{-j2\pi \frac{(2)(1)}{4}} + x[2] e^{-j2\pi \frac{(2)(2)}{4}} + x[3] e^{-j2\pi \frac{(2)(3)}{4}}$$

$$x[0] + x[1] e^{-j\pi} + x[2] e^{-j2\pi} + x[3] e^{-j3\pi}$$

$$2 + (3)(-1) + (-1)(1) + (1)(-1)$$

$$2 - 3 - 1 - 1$$

$$-3$$

$$x[3] = \sum_{n=0}^3 x[n] e^{-j2\pi \frac{3n}{4}}$$

$$x[0] e^{-j2\pi \frac{(3)(0)}{4}} + x[1] e^{-j2\pi \frac{(3)(1)}{4}} + x[2] e^{-j2\pi \frac{(3)(2)}{4}} + x[3] e^{-j2\pi \frac{(3)(3)}{4}}$$

$$x[0] + x[1] e^{-j\frac{3\pi}{2}} + x[2] e^{-j3\pi} + x[3] e^{-j\frac{9\pi}{2}}$$

$$(2) + (3)(j) + (-1)(-1) + (1)(-j)$$

$$2 + 3j + 1 - j$$

$$3 + 2j$$

$$\underline{x[k] = \{5, 3 - 2j, 3, 3 + 2j\}}$$

Final Answer



DISCRETE FOURIER TRANSFORM (CONT.)



$$\begin{bmatrix} x[0] \\ x[1] \\ x[2] \\ x[3] \end{bmatrix} = \begin{bmatrix} w_0^4 & w_0^4 & w_0^4 & w_0^4 \\ w_0^4 & w_1^4 & w_2^4 & w_3^4 \\ w_0^4 & w_2^4 & w_4^4 & w_6^4 \\ w_0^4 & w_3^4 & w_4^4 & w_9^4 \end{bmatrix} \begin{bmatrix} x[0] \\ x[1] \\ x[2] \\ x[3] \end{bmatrix}$$

$$w_n = e^{-j\frac{2\pi n}{N}} \quad w_3 = e^{-j\frac{2\pi(3)}{4}} \quad w_9 = e^{-j\frac{2\pi(9)}{4}}$$

$$w_0 = e^{-j\frac{2\pi(0)}{4}} \quad w_3 = e^{-j\frac{2\pi}{2}} \quad w_9 = e^{-j\frac{9\pi}{2}}$$

$$w_0 = 1 \quad w_3 = j \quad w_9 = -j$$

$$w_1 = e^{-j\frac{2\pi(1)}{4}} \quad w_4 = e^{-j\frac{2\pi(4)}{4}}$$

$$w_1 = e^{-j\frac{\pi}{2}} \quad w_4 = e^{-j2\pi}$$

$$w_1 = -j \quad w_4 = 1$$

$$w_2 = e^{-j\frac{2\pi(2)}{4}} \quad w_6 = e^{-j\frac{2\pi(6)}{4}}$$

$$w_2 = e^{-j\pi} \quad w_6 = e^{-j3\pi}$$

$$w_2 = -1 \quad w_6 = -1$$

$$\begin{bmatrix} x[0] \\ x[1] \\ x[2] \\ x[3] \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} 2 \\ 3 \\ -1 \\ 1 \end{bmatrix}$$

$$x[0] = (1)(2) + (1)(3) + (1)(-1) + (1)(1)$$

$$x[0] = 2 + 3 - 1 + 1 = 5$$

$$x[1] = (1)(2) + (j)(3) + (-1)(-1) + (j)(1)$$

$$x[1] = 2 + 3j + 1 + j = 3 - 2j$$

$$x[2] = (1)(2) + (-1)(3) + (1)(-1) + (-1)(1)$$

$$x[2] = 2 - 3 - 1 - 1 = -3$$

$$x[3] = (1)(2) + (j)(3) + (-1)(-1) + (-j)(1)$$

$$x[3] = 2 + 3j + 1 - j = 3 + 2j$$

$$x[k] = \{5, 3 - 2j, -3, 3 + 2j\}$$



INVERSE DFT



$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} x[k] e^{j2\pi \frac{k(n)}{N}}$$

$$x[k] = [5, 3 - 2j, -3, 3 + 2j]$$

$$x[0] = \frac{1}{4} \sum_{k=0}^3 x[k] e^{j2\pi \frac{k(0)}{4}}$$

$$x[0] = \frac{1}{4} \left[x[0] e^{j2\pi \frac{(0)(0)}{4}} + x[1] e^{j2\pi \frac{(1)(0)}{4}} + x[2] e^{j2\pi \frac{(2)(0)}{4}} + x[3] e^{j2\pi \frac{(3)(0)}{4}} \right]$$

$$x[0] = \frac{1}{4} [x[0] + x[1] + x[2] + x[3]]$$

$$x[0] = \frac{1}{4} [5 + (3 - 2j) + (-3) + (3 + 2j)]$$

$$x[0] = \frac{1}{4} [5 + (3 - 2j) + (-3) + (3 + 2j)]$$

$$x[0] = \frac{1}{4} [5 + 3] = \frac{1}{4} (8) = 2$$

$$x[1] = \frac{1}{4} \left[x[0] e^{j2\pi \frac{(0)(1)}{4}} + x[1] e^{j2\pi \frac{(1)(1)}{4}} + x[2] e^{j2\pi \frac{(2)(1)}{4}} + x[3] e^{j2\pi \frac{(3)(1)}{4}} \right]$$

$$x[1] = \frac{1}{4} \left[x[0] + x[1] e^{j\frac{\pi}{2}} + x[2] e^{j\pi} + x[3] e^{j\frac{3\pi}{2}} \right]$$

$$x[1] = \frac{1}{4} [5 + (3 - 2j)(j) + (-3)(-1) + (3 + 2j)(-j)]$$

$$x[1] = \frac{1}{4} [5 + (3j - 2j^2) + 3 - (-3 - 2j^2)]$$

$$x[1] = \frac{1}{4} [8 - 4j^2]$$

$$x[1] = \frac{1}{4} [8 - 4(-1)] = \frac{1}{4} [8 + 4] = \frac{1}{4} (12) = 3$$

$$x[2] = \frac{1}{4} \left[x[0] e^{j2\pi \frac{(0)(2)}{4}} + x[1] e^{j2\pi \frac{(1)(2)}{4}} + x[2] e^{j2\pi \frac{(2)(2)}{4}} + x[3] e^{j2\pi \frac{(3)(2)}{4}} \right]$$

$$x[2] = \frac{1}{4} [x[0] + x[1] e^{j\pi} + x[2] e^{j2\pi} + x[3] e^{j3\pi}]$$

$$x[2] = \frac{1}{4} [5 + (3 - 2j)(-1) + (-3)(1) + (3 + 2j)(-1)]$$

$$x[2] = \frac{1}{4} [5 + (3 + 2j) + (-3) + (-3 - 2j)]$$

$$x[2] = \frac{1}{4} [5 + (-3) + (-3) + (-3)]$$

$$x[2] = \frac{1}{4} [5 - 9]$$

$$x[2] = \frac{1}{4} [-4] = -1$$

INVERSE DFT (CONT.)



$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} x[k] e^{j2\pi \frac{k(n)}{N}}$$

$$x[k] = [5, 3 - 2j, -3, 3 + 2j]$$

$$x[3] = \frac{1}{4} \left[x[0] e^{j2\pi \frac{(0)(3)}{4}} + x[1] e^{j2\pi \frac{(1)(3)}{4}} + x[2] e^{j2\pi \frac{(2)(3)}{4}} + x[3] e^{j2\pi \frac{(3)(3)}{4}} \right]$$

$$x[3] = \frac{1}{4} \left[x[0] + x[1] e^{\frac{3\pi}{2}} + x[2] e^{3\pi} + x[3] e^{j \frac{9\pi}{2}} \right]$$

$$x[3] = \frac{1}{4} [5 + (3 - 2j)(-j) + (-3)(-1) + (3 + 2j)(j)]$$

$$x[3] = \frac{1}{4} [5 + (-3 - 2j^2) + (-3) + (3j + 2j^2)]$$

$$x[3] = \frac{1}{4} [8 + 4j^2]$$

$$x[3] = \frac{1}{4} [8 + 4(-1)] = \frac{1}{4} [8 - 4] = \frac{4}{4} = 1$$

FOURIER SERIES



- A Fourier series is an expansion of a periodic function $f(x)$ in terms of an infinite sum of sines and cosines
- Fourier series make use of the orthogonality relationships of the sine and cosine functions
- The computation and study of Fourier series is known as harmonic analysis and is extremely useful to break up an arbitrary periodic function into a set of simple terms that can be plugged in, solved individually, and then recombined to obtain the solution to the original problem or an approximation to it to whatever accuracy is desired or practical

Joseph Fourier, in full Jean-Baptiste-Joseph, Baron Fourier, (born March 21, 1768, Auxerre, France—died May 16, 1830, Paris)



FOURIER SERIES WITH PERIOD 2π

(ii) Let $f(x)$ be periodic with period 2π and piecewise continuous in the interval $[-\pi, \pi]$. Then the Fourier series of $f(x)$ is given by

$$f(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

with coefficients

$$\left. \begin{aligned} a_0 &= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) \, dx, \\ a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx \, dx, \quad n = 1, 2, 3, \dots \\ b_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx \, dx, \quad n = 1, 2, 3, \dots \end{aligned} \right\}$$

EVEN & ODD FUNCTIONS

Fourier series takes on simpler forms for Even and Odd functions

Even function



A function is Even if $f(x) = f(-x)$ for all x .
The graph of an even function is symmetric about the y-axis.

Odd function



A function is Odd if $f(x) = -f(-x)$ for all x .
The graph of an even function is skew-symmetric about the y-axis.

Example 1



- Let $f(x)$ be a function of period 2π such that $f(x) = x/2$ $[0, 2\pi]$.
- Find the Fourier series of $f(x)$ over $[0, 2\pi]$

$$p = 2\pi$$

$$f(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

$$a_0 = \frac{1}{2\pi} \int_0^{2\pi} \frac{x}{2} dx$$

$$a_0 = \frac{1}{4\pi} \int_0^{2\pi} x dx$$

$$a_0 = \frac{1}{4\pi} \left(\frac{x^2}{2} \right) \Big|_0^{2\pi}$$

$$a_0 = \frac{1}{4\pi} \left[\frac{(2\pi)^2}{2} - \frac{(0)^2}{2} \right]$$

$$a_0 = \frac{1}{4\pi} \left(\frac{4\pi^2}{2} \right)$$

$$a_0 = \frac{\pi}{2}$$

$$a_n = \frac{1}{\pi} \int_0^{2\pi} \left(\frac{x}{2} \right) \cos nx dx$$

$$a_n = \frac{1}{2\pi} \int_0^{2\pi} x \cos nx dx$$

$$\int u dv = uv - \int v du$$

$$u = x \quad dv = \cos nx dx$$

$$du = dx \quad v = \frac{1}{n} \sin nx$$

$$a_n = \frac{1}{2\pi} \left\{ \left[\frac{x}{n} \sin nx \right] \Big|_0^{2\pi} - \frac{1}{n} \int_0^{2\pi} \sin nx \right\}$$

$$a_n = \frac{1}{2\pi} \left\{ \left[\frac{1}{n} (2\pi \sin 2\pi n - 0 \sin(0)) \right] + \frac{1}{n^2} (\cos nx) \Big|_0^{2\pi} \right\}$$

$$a_n = \frac{1}{2\pi} \left[\frac{1}{n^2} (\cos 2\pi n - \cos n(0)) \right]$$

$$a_n = \frac{1}{2\pi} \left[\frac{1}{n^2} (1 - 1) \right]$$

$$a_n = \frac{1}{2\pi} \left[\frac{1}{n^2} (0) \right]$$

$$a_n = 0$$

$$b_n = \frac{1}{\pi} \int_0^{2\pi} \frac{x}{2} \sin nx dx$$

$$b_n = \frac{1}{2\pi} \int_0^{2\pi} x \sin x dx$$

$$u = x \quad dv = \sin nx dx$$

$$du = dx \quad v = -\frac{1}{n} \cos nx$$

$$b_n = \frac{1}{2\pi} \left[\left(-\frac{x}{n} \cos nx \right) \Big|_0^{2\pi} + \frac{1}{n} \int_0^{2\pi} \cos nx dx \right]$$

$$b_n = \frac{1}{2\pi} \left[\left(-\frac{x}{n} \cos nx \right) \Big|_0^{2\pi} + \left(\frac{1}{n^2} \sin nx \right) \Big|_0^{2\pi} \right]$$

$$b_n = \frac{1}{2\pi} \left[-\frac{1}{n} (2\pi \cos 2\pi n - 0 \cos n(0)) + \frac{1}{n^2} (\sin 2\pi - \sin n(0)) \right]$$

$$b_n = \frac{1}{2\pi} \left[-\frac{1}{n} (2\pi) \right]$$

$$b_n = -\frac{1}{n}$$

$$f(x) = \frac{\pi}{2} - \frac{1}{n} \sum_{n=1}^{\infty} \sin nx$$

Example 2



$$f(x) = \begin{cases} k, & 0 < x < \pi \\ -k, & -\pi < x < 0 \end{cases}$$

$$f(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

$$a_0 = \frac{1}{2\pi} \left[\int_0^{\pi} k dx + \int_{-\pi}^0 -k dx \right]$$

$$a_0 = \frac{1}{2\pi} [(kx)|_0^{\pi} + (-kx)|_{-\pi}^0]$$

$$a_0 = [k\pi - k(0)] - k(0) + k(\pi)$$

$$a_0 = \frac{1}{2\pi} [k\pi - k\pi]$$

$$a_0 = 0$$

$$a_n = \frac{1}{\pi} \left[\int_0^{\pi} k \cos nx dx + \int_{-\pi}^0 -k \cos nx dx \right]$$

$$a_n = \frac{1}{\pi} \left[\frac{k}{n} (\sin nx)|_0^{\pi} - \frac{k}{n} (\sin nx)|_{-\pi}^0 \right]$$

$$a_n = \frac{1}{\pi} \left[\frac{k}{n} (\sin n\pi - \sin n(0)) - \frac{k}{n} (\sin n(0) + \sin n\pi) \right]$$

$$a_n = \frac{1}{\pi} \left[\frac{k}{n} - \frac{k}{n} \right]$$

$$a_n = \frac{1}{\pi} [0]$$

$$a_n = 0$$

$$b_n = \frac{1}{\pi} \left[\int_0^{\pi} k \sin nx dx + \int_{-\pi}^0 -k \sin nx dx \right]$$

$$b_n = \frac{1}{\pi} \left[-\frac{k}{n} (\cos nx)|_0^{\pi} + \frac{k}{n} (\cos nx)|_{-\pi}^0 \right]$$

$$b_n = \frac{1}{\pi} \left[-\frac{k}{n} (\cos n\pi - \cos n(0)) + \frac{k}{n} (\cos n(0) - \cos n\pi) \right]$$

$$b_n = -\frac{k}{\pi n} (-\cos n\pi + 1 + 1 - \cos n\pi)$$

$$b_n = \frac{2k}{\pi n} (1 - \cos n\pi)$$

$$f(x) = \sum_{n=1}^{\infty} \frac{2k}{\pi n} (1 - \cos n\pi) \sin nx$$

$$f(x) = \frac{2k}{\pi} \sum_{n=1}^{\infty} \frac{(1 - \cos n\pi)}{n} \sin nx$$

Example 3



$$f(x) = \begin{cases} 1, & -\frac{\pi}{2} < x < \frac{\pi}{2} \\ -1, & \frac{\pi}{2} < x < \frac{3\pi}{2} \end{cases}$$

$$f(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

$$a_0 = \frac{1}{2\pi} \left[\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} 1 dx + \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} -1 dx \right]$$

$$a_0 = \frac{1}{2\pi} \left[\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} x dx + \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} -x dx \right]$$

$$a_0 = \frac{1}{2\pi} \left\{ \left(\frac{\pi}{2} - \left(-\frac{\pi}{2} \right) \right) + \left(-\frac{\pi}{2} + \frac{3\pi}{2} \right) \right\}$$

$$a_0 = \frac{1}{2\pi} \left[\frac{2\pi}{2} + \left(-\frac{2\pi}{2} \right) \right]$$

$$a_0 = \frac{1}{2\pi} [0]$$

$$a_0 = 0$$

$$a_n = \frac{1}{\pi} \left[\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (1) \cos nx dx + \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} (-1) \cos nx dx \right]$$

$$a_n = \frac{1}{\pi} \left[\frac{1}{\pi} (\sin nx) \Big|_{-\frac{\pi}{2}}^{\frac{\pi}{2}} - \frac{1}{\pi} (\sin nx) \Big|_{\frac{\pi}{2}}^{\frac{3\pi}{2}} \right]$$

$$a_n = \frac{1}{\pi} \left[\frac{1}{\pi} (1 - (-1)) - \frac{1}{\pi} (-1 - 1) \right]$$

$$a_n = \frac{1}{\pi} \left[\frac{1}{\pi} (2) - \frac{1}{\pi} (-2) \right]$$

$$a_n = \frac{1}{\pi} \left(\frac{4}{\pi} \right)$$

$$a_n = \frac{4}{\pi n}$$

$$b_n = \frac{1}{\pi} \left(\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin nx dx + \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} -\sin nx dx \right)$$

$$b_n = \frac{1}{\pi} \left[-\frac{1}{\pi} (\cos nx) \Big|_{-\frac{\pi}{2}}^{\frac{\pi}{2}} + \frac{1}{\pi} (\cos nx) \Big|_{\frac{\pi}{2}}^{\frac{3\pi}{2}} \right]$$

$$b_n = \frac{1}{\pi} \left[-\frac{1}{\pi} (0) + \frac{1}{\pi} (0) \right]$$

$$b_n = \frac{1}{\pi} [0]$$

$$b_n = 0$$

$$f(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

$$f(x) = \frac{4}{\pi n} \sum_{n=1}^{\infty} \cos nx$$

$$f(x) = \frac{4}{\pi} \cos x + \frac{2}{\pi} \cos 2x + \frac{4x}{3\pi} \cos 3x + \frac{1}{\pi} \cos 4x$$

DISCRETE TIME FOURIER TRANSFORM



$$x[e^{j\omega}] = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$$



$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} x(e^{j\omega})(e^{j\omega n}) d\omega$$



$$\begin{aligned}\delta^n &= 1 \\ \delta(n-1) &= e^{-j\omega} \times 1 \\ \delta(n-2) &= e^{-j2\omega} \times 1 \\ \delta(n+2) &= e^{j2\omega} \times 1\end{aligned}$$

$$\text{Goal: } \sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$$



$$\begin{aligned}x(n) &= a^n u(n); a < 1 \\ e^{j\omega} &= \sum_{n=-\infty}^{\infty} \delta[n]e^{-j\omega n(0)} \\ e^{j\omega} &= \sum_{n=-\infty}^{\infty} a^n u[n]e^{-j\omega n} \\ x[e^{j\omega}] &= \sum_{n=0}^{\infty} a^n (1)e^{-j\omega n} \\ x[e^{j\omega}] &= \sum_{n=0}^{\infty} (a e^{-j\omega})^n\end{aligned}$$

$$\frac{1}{1 - \frac{e^{j\omega}}{ae^{-j\omega}}}$$



$$\begin{aligned}x(n) &= 2^n u(-n) \\ x[e^{j\omega}] &= \sum_{n=-\infty}^{\infty} 2^n u[-n]e^{-j\omega n} \\ x[e^{j\omega}] &= \sum_{n=-\infty}^0 2^n (1)e^{-j\omega n} \\ x[e^{j\omega}] &= \sum_{n=0}^{\infty} 2^{-n} e^{j\omega n} \\ x[e^{j\omega}] &= \sum_{n=0}^{\infty} (2^{-1} e^{j\omega})^n \\ x[e^{j\omega}] &= \sum_{n=0}^{\infty} \left(\frac{e^{j\omega}}{2}\right)^n\end{aligned}$$

$$\frac{1}{1 - \frac{e^{j\omega}}{2}}$$



$$\begin{aligned}x(n) &= \left(\frac{1}{3}\right)^n u(n-3) \\ x[e^{j\omega}] &= \sum_{n=-\infty}^{\infty} \left(\frac{1}{3}\right)^n u[n-3]e^{-j\omega n} \\ x[e^{j\omega}] &= \sum_{n=3}^{\infty} \left(\frac{1}{3}\right)^n (1)e^{-j\omega n} \\ m &= n-3 \\ n &= m+3 \\ x[e^{j\omega}] &= \sum_{m=0}^{\infty} \left(\frac{1}{3}\right)^{m+3} e^{-j\omega(m+3)} \\ x[e^{j\omega}] &= \sum_{m=0}^{\infty} \left(\frac{1}{3}\right)^3 \left(\frac{1}{3}\right)^m e^{-j3\omega} e^{-j\omega m} \\ \left(\frac{1}{27}\right)(e^{-j3\omega}) &= \sum_{m=0}^{\infty} \left(\frac{e^{j\omega}}{3}\right)^m \\ \left(\frac{e^{-j3\omega}}{27}\right)\left(\frac{1}{1 - \frac{e^{-j\omega}}{3}}\right)\end{aligned}$$

$$\left(\frac{e^{-j3\omega}}{27\left(1 - \frac{e^{-j\omega}}{3}\right)}\right)$$



$$x(n) = \delta[n] = 1$$

DISCRETE TIME FOURIER TRANSFORM



$$1. \quad x(e^{j\omega}) = \frac{6 - \frac{2}{3}e^{-j\omega} - \frac{1}{6}e^{-2j\omega}}{-\frac{1}{6}e^{-j2\omega} + \frac{1}{6}e^{-j\omega} + 1}$$

Find $x(n)$

$$(e^{j\omega})^2 = (t)^2$$

$$x(e^{j\omega}) = \frac{6 - \frac{2}{3}t - \frac{1}{6}t^2}{-\frac{1}{6}t^2 + \frac{1}{6}t + 1}$$

$$x(e^{j\omega}) = \frac{-\frac{1}{6}t^2 - \frac{2}{3}t + 6}{-\frac{1}{6}t^2 + \frac{1}{6}t + 1} * \frac{-6}{-6}$$

$$x(e^{j\omega}) = \frac{t^2 + 4t - 36}{t^2 - t - 6}$$

$$1 + \frac{5t - 30}{t^2 - t - 6}$$

$$1 + \frac{5t - 30}{(t - 3)(t + 2)}$$

$$= \frac{A}{t - 3} + \frac{B}{t + 2}$$

$$5t - 30 = A(t + 2) + B(t - 3)$$

$$5 = A + B \rightarrow 15 = 3A + 3B$$

$$-30 = 2A + 2B \rightarrow -30 = 2A + 3B$$

$$-15 = 5A$$

$$A = -3$$

$$B = 5 - A$$

$$B = 5 - (-3)$$

$$B = 8$$

$$= 1 + \frac{-3}{t - 3} + \frac{8}{t + 2}$$

$$= 1 + \frac{-3}{-(3 - t)} + \frac{8}{2 - (-t)}$$

$$= 1 + \frac{-3}{-3 - t} * \frac{1}{\frac{1}{3}} + \frac{8}{2 - (-t)} * \frac{1}{\frac{1}{2}}$$

$$= 1 + \frac{1}{1 - \frac{t}{3}} + 4 \left(\frac{1}{1 - \left(-\frac{t}{2}\right)} \right)$$

$$= 1 + \frac{1}{1 - e^{-j\omega} \frac{1}{3}} + 4 \left(\frac{1}{1 - \frac{-e^{-j\omega}}{2}} \right)$$

$$\delta[n] + \left(\frac{1}{3}\right)^n u(n) + 2/\left(-\frac{1}{2}\right)^n u[n]$$

$$2. \quad x(e^{j\omega}) = \frac{3 - \left(\frac{1}{4}\right)e^{-j\omega}}{-\frac{1}{16}e^{-2j\omega} + 1}$$

find (x)

$$x(e^{j\omega}) = \frac{3 - \frac{1}{4}t}{-\frac{1}{16}t^2 + 1}$$

$$= \frac{-\frac{1}{4}t + 3}{-\frac{1}{16}t^2 + 1} * \frac{16}{16}$$

$$= \frac{4t - 48}{t^2 - 16}$$

$$= \frac{A}{t - 4} + \frac{B}{t + 4}$$

$$4t - 48 = A(t + 4) + B(t - 4)$$

$$4 = A + B \rightarrow 16 = 4A + 4B$$

$$-48 = 4A - 4B \rightarrow -48 = 4A - 4B$$

$$-32 = 8A$$

$$A = -4$$

$$B = 4 - (4)$$

$$B = 8$$

$$= \frac{-4}{t - 4} + \frac{8}{t + 4}$$

$$= \frac{-4}{-(4 - t)} * \frac{1}{\frac{1}{4}} + \frac{8}{4 - (-t)} * \frac{1}{\frac{1}{4}}$$

$$= \frac{1}{1 - \frac{t}{4}} + 2 \left(\frac{1}{1 - \left(-\frac{t}{4}\right)} \right)$$

$$= \left(\frac{1}{4}\right)^n u[n] + 2 \left(\frac{1}{4}\right)^n u[n]$$

$$3. \quad x[e^{j\omega}] = e^{j\omega}$$

$$\delta[n+1]$$

$$e^{j\omega} = \cos \omega + j \sin \omega$$

$$e^{-j\omega} = \cos \omega - j \sin \omega$$

$$\frac{e^{j\omega} + e^{-j\omega}}{2} = \frac{2 \cos \omega}{2}$$

$$\cos \omega = \frac{e^{j\omega} + e^{-j\omega}}{2}$$

$$= \cos 2\omega = \frac{e^{j2\omega} + e^{-j2\omega}}{2}$$

$$4. \quad x(e^{j\omega}) = \cos^2 \omega$$

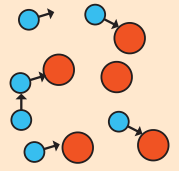
$$x(e^{j\omega}) = \frac{(1 + \cos 2\omega)}{2}$$

$$x(e^{j\omega}) = \frac{1}{2} + \frac{1}{2} \cos 2\omega$$

$$= \frac{1}{2} + \frac{1}{4} (e^{j2\omega} + e^{-j2\omega})$$

$$= \frac{1}{2} \delta[n] + \frac{1}{4} [\delta[n + 2] + \delta[n - 2]]$$

CONVOLUTION



Perform the following on the sequence:

$$x(n) = \{1, 2, 3, 1\}$$

$$h(n) = \{1, 1, 1\}$$

a. Linear Convolution

$$\begin{array}{r} 1 \ 2 \ 3 \ 1 \\ \underline{1 \ 1 \ 1} \end{array}$$

$$1 \ 2 \ 3 \ 1$$

$$1 \ 2 \ 3 \ 1 \ x$$

$$\underline{1 \ 2 \ 3 \ 1 \ x \ x}$$

$$1 \ 3 \ 6 \ 6 \ 4 \ 1$$

$$y[n] = \{1, 3, 6, 6, 4, 1\}$$

Determine the linear convolution of the following system signals

$$x(n) = \delta(n) + 3\delta(n-1) + 2\delta(n-1) - \delta(n-3) + \delta(n-4)$$

$$h(n) = 2\delta(n) - \delta(n-1)$$

$$x(n) = \{1, 3, 2, -1, 1\}$$

$$h(n) = \{2, -1\}$$

$$\begin{array}{r} 1 \ 3 \ 2 \ -1 \ 1 \\ \underline{2 \ -1} \end{array}$$

$$2 \ -1$$

$$-1 \ -3 \ -2 \ 1 \ -1$$

$$\underline{2 \ 6 \ 4 \ -2 \ 2 \ x}$$

$$2 \ 5 \ -1 \ -4 \ 3 \ -1$$

b. Circular Convolution

$$\begin{bmatrix} 1 & 1 & 3 & 2 \\ 2 & 1 & 1 & 3 \\ 3 & 2 & 1 & 1 \\ 1 & 3 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 5 \\ 4 \\ 6 \\ 6 \end{bmatrix}$$

$$y(n) = \{2, 5, -1, -4, 3, -1\}$$

$$(1)(1) + (1)(1) + (3)(1) + (2)(0) = 5$$

$$(2)(1) + (1)(1) + (1)(1) + (3)(0) = 4$$

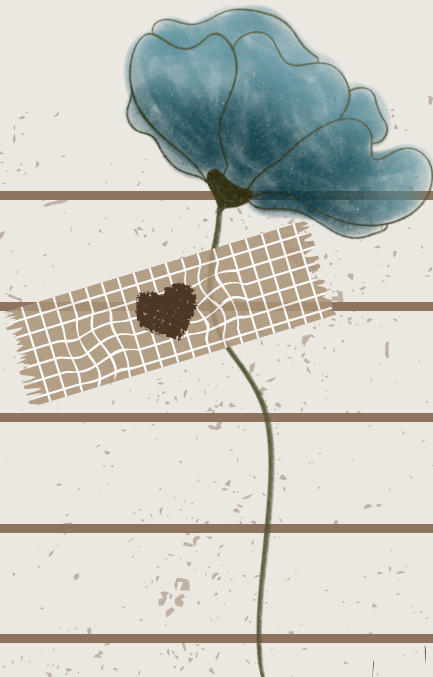
$$(3)(1) + (2)(1) + (1)(1) + (1)(0) = 6$$

$$(1)(1) + (3)(1) + (1)(1) + (1)(0) = 6$$

$$y[n] = \{5, 4, 6, 6\}$$

c. Linear Convolutional using Circular Convolutional

$$\begin{bmatrix} 1 & 0 & 0 & 1 & 3 & 2 \\ 2 & 1 & 0 & 0 & 1 & 3 \\ 3 & 2 & 1 & 0 & 0 & 1 \\ 1 & 3 & 2 & 1 & 0 & 0 \\ 0 & 1 & 3 & 2 & 1 & 0 \\ 0 & 0 & 1 & 3 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 6 \\ 6 \\ 4 \\ 1 \end{bmatrix}$$



DiGiTAL **SiGNA** **P R o C E S S I N G** **F i N A L s**

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Z-TRANSFORM

- The z-transform of a discrete time signal $x(n)$ is defined as the power series

$$X(z) = \sum_{n=-\infty}^{\infty} x(n)z^{-n}$$

- Where z is a complex variable. It is sometimes called the direct z-transform because it transforms the time domain signal $x(n)$ into its complex-plane representation $X(z)$.
- The ROC (region of convergence) of $X(z)$ is the set of all values of z for which $X(z)$ attains a finite value.

Z-transform



$$X(z) = \sum_{n=-\infty}^{\infty} x(n)z^{-n}$$

$z = re^{j\theta}$ - polar form of complex variable z

r - magnitude of the complex variable z

$r = |z| = \text{mod } z$

$\theta = \text{complex argument or phase or angle}$

Examples



1 $x[n] = \left(\frac{1}{2}\right)^n u[n]$

$$\begin{aligned} x[z] &= \sum_{n=-\infty}^{\infty} x[n]z^{-n} \\ &= \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n (1)z^{-n} \\ &= \sum_{n=0}^{\infty} \left(\frac{1}{2}z^{-1}\right)^n, \end{aligned}$$

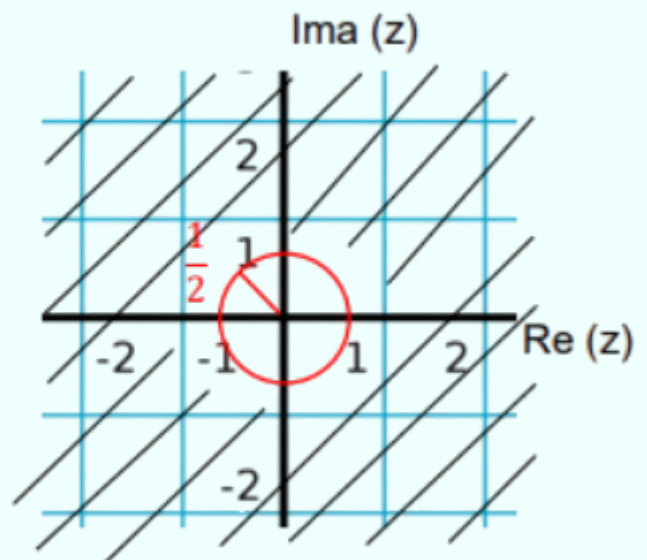
$$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$$

ROC:

$$\left|\frac{1}{2}z^{-1}\right| < 1$$

$$\left|\frac{1}{2z}\right| < 1$$

$$|z| > \frac{1}{2}$$



$$x[z] = \frac{1}{1 - \frac{1}{2}z^{-1}}$$

2

$$x[n] = u[n]$$

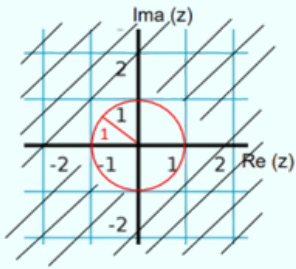
$$= \sum_{n=-\infty}^{\infty} (1)z^{-n}$$

$$= \sum_{n=0}^{\infty} (z^{-1})^n,$$

$$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$$

$$x[z] = \frac{1}{1-z^{-1}}$$

ROC:
 $|z^{-1}| < 1$
 $\left|\frac{1}{z}\right| < 1$
 $|z| > 1$



3

$$x[n] = -u[-n-1]$$

$$x[z] = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$$

$$= \sum_{n=-\infty}^{-1} (-1)z^{-n}$$

$$= - \sum_{n=1}^{\infty} (z)^n, \quad n = -n$$

$$= - \sum_{m=0}^{\infty} (z)^{m+1}, \quad \begin{matrix} n-1 = m \\ n = m+1 \end{matrix}$$

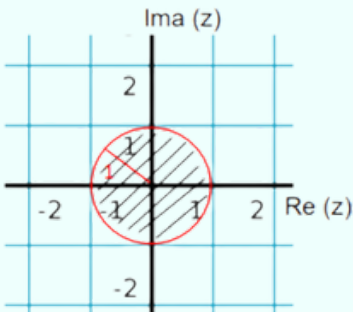
$$= - \sum_{m=0}^{\infty} z^m z$$

$$= -z \sum_{m=0}^{\infty} z^m, \quad \sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$$

$$= \frac{-z}{1-z} = \left[\frac{-z}{1-z}\right] \frac{-z}{-z}$$

$$x[z] = \frac{1}{1-\frac{1}{z}}$$

ROC:
 $|z^{-1}| < 1$
 $\left|\frac{1}{z}\right| < 1$
 $|z| < 1$



4

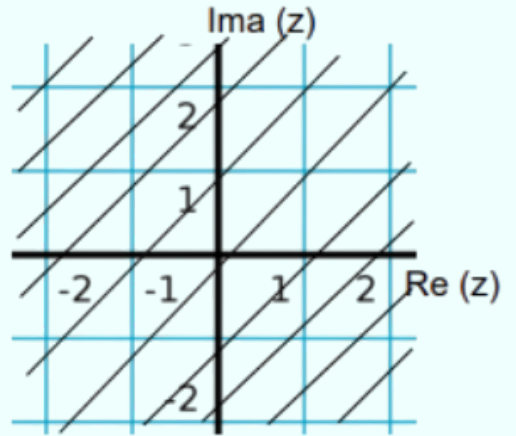
$$x[n] = \delta[n]$$

$$x[z] = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$$

$$= \sum_{n=-\infty}^{-1} \delta[n]z^{-n}$$

$$= z^{-n}|_{n=0} = 1$$

$$x[z] = z^0 = 1$$



5

$$x[n] = \delta[n-k]$$

$$x[z] = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$$

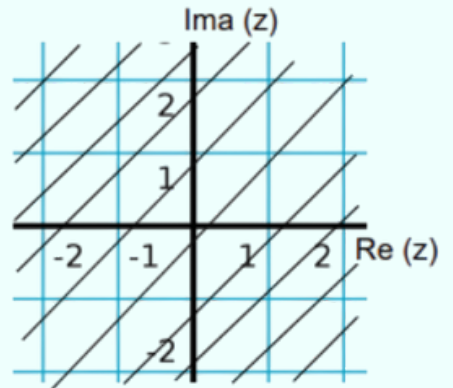
$$= \sum_{n=-\infty}^{-1} \delta[n]z^{-n}$$

$$= z^{-n}|_{n=k}$$

$$x[z] = z^{-k}$$

$z = 0$ when $k > 0$

$z = \infty$ when $k < 0$

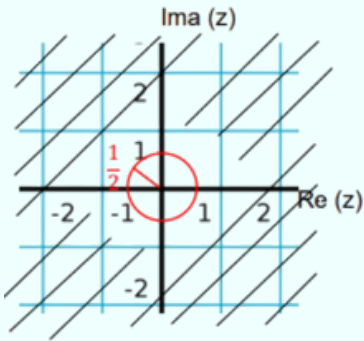


6

$$\begin{aligned}
 x[n] &= 7\left(\frac{1}{3}\right)^n - 6\left(\frac{1}{2}\right)^n u[n] \\
 x[z] &= \sum_{n=-\infty}^{\infty} x[n]z^{-n} \\
 &= \sum_{n=0}^{\infty} \left[7\left(\frac{1}{3}\right)^n - 6\left(\frac{1}{2}\right)^n \right] z^{-n} \\
 &= 7 \sum_{n=0}^{\infty} \left(\frac{1}{3}\right)^n z^{-n} - 6 \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n z^{-n} \\
 &= 7 \sum_{n=0}^{\infty} \left(\frac{1}{3}z^{-1}\right)^n - 6 \sum_{n=0}^{\infty} \left(\frac{1}{2}z^{-1}\right)^n, \quad \sum_{n=0}^{\infty} x^n = \frac{1}{1-x}
 \end{aligned}$$

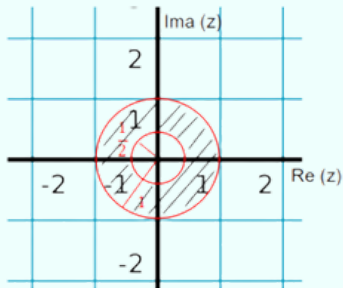
$$x[z] = \frac{7}{1 - \frac{1}{3}z^{-1}} - \frac{6}{1 - \frac{1}{2}z^{-1}}$$

$$\begin{aligned}
 \text{ROC}_1: \quad & \left| \frac{1}{3}z^{-1} \right| < 1 \quad \left| \frac{1}{2}z^{-1} \right| < 1 \\
 & \left| \frac{1}{3z} \right| < 1 \quad \left| \frac{1}{2z} \right| < 1 \\
 & |z| > \frac{1}{3} \quad |z| > \frac{1}{2}
 \end{aligned}$$



$$\begin{aligned}
 \text{ROC}_1: \quad & |z^{-1}| < 1 \\
 & \left| \frac{1}{z} \right| < 1 \\
 & |z| < 1 \\
 \text{ROC}_2: \quad & \left| \frac{1}{2}z^{-1} \right| < 1 \\
 & \left| \frac{1}{2z} \right| < 1 \\
 & |z| > \frac{1}{2}
 \end{aligned}$$

$$\text{ROC: } \frac{1}{2} < |z| < 1$$



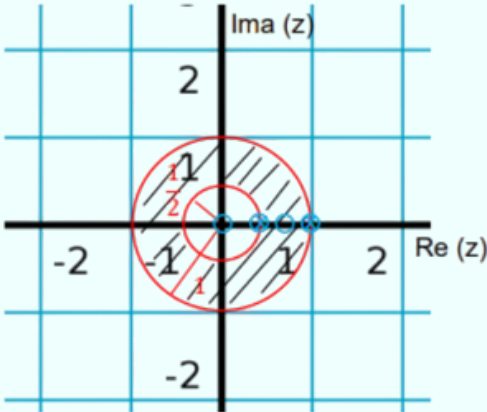
$$\begin{aligned}
 x[z] &= \left[\frac{1}{1 - \frac{1}{3}z^{-1}} + \frac{1}{1 - \frac{1}{2}z^{-1}} \right] \frac{z}{z} \\
 &= \frac{z}{z-1} + \frac{z}{z-\frac{1}{2}} \\
 &= \frac{z(z-\frac{1}{2}) + z(z-1)}{(z-1)(z-\frac{1}{2})} \\
 &= \frac{z^2 - \frac{1}{2}z + z^2 - z}{(z-1)(z-\frac{1}{2})} \\
 &= \frac{2z^2 - \frac{3}{2}z}{(z-1)(z-\frac{1}{2})} \\
 &= \frac{z(2z - \frac{3}{2})}{(z-1)(z-\frac{1}{2})}
 \end{aligned}$$

7

$$\begin{aligned}
 x[n] &= -u[-n-1] + \left(\frac{1}{2}\right)^n u[n] \\
 x[z] &= \sum_{n=-\infty}^{\infty} x[n]z^{-n} \\
 &= \sum_{n=-\infty}^{\infty} \left[-u[-n-1] + \left(\frac{1}{2}\right)^n u[n] \right] z^{-n} \\
 &= - \sum_{n=-\infty}^{-1} z^{-n} + \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n z^{-n}, \quad n = -n \\
 &= - \sum_{m=0}^{\infty} z^m + \sum_{n=0}^{\infty} \left(\frac{1}{2}z^{-1}\right)^n, \quad n-1=m, \quad n=m+1 \\
 &= - \sum_{m=0}^{\infty} z^{m+1} + \sum_{n=0}^{\infty} \left(\frac{1}{2}z^{-1}\right)^n \\
 &= - \sum_{m=0}^{\infty} z^m z + \sum_{n=0}^{\infty} \left(\frac{1}{2}z^{-1}\right)^n \\
 &= -z \sum_{m=0}^{\infty} z^m + \sum_{n=0}^{\infty} \left(\frac{1}{2}z^{-1}\right)^n, \quad \sum_{n=0}^{\infty} x^n = \frac{1}{1-x} \\
 &= \frac{-z}{1-z} + \frac{1}{1-\frac{1}{2}z^{-1}} = \frac{-z}{-z} \left[\frac{-z}{1-z} \right] + \frac{1}{1-\frac{1}{2}z^{-1}}
 \end{aligned}$$

$$x[z] = \frac{1}{1-z} + \frac{1}{1-\frac{1}{2}z^{-1}}$$

$$\begin{aligned}
 \text{Zeroes} \quad & z = 0 \\
 & 2z - \frac{3}{2} = 0 \\
 & z = \frac{3}{4} \\
 \text{Poles} \quad & z - 1 = 0 \\
 & z = 1 \\
 & z - \frac{1}{2} = 0 \\
 & z = \frac{1}{2}
 \end{aligned}$$



8

$$x[n] = \left(\frac{1}{3}\right)^n \sin\left(\frac{\pi n}{4}\right) u[n]$$

$$= \sum_{n=0}^{\infty} \left[\left(\frac{1}{3}\right)^n \sin\left(\frac{\pi n}{4}\right) u[n] \right] z^{-n}, \quad \sin \theta = \frac{e^{j\theta} - e^{-j\theta}}{2j}$$

$$= \sum_{n=0}^{\infty} \left(\frac{1}{3}\right)^n \left(\frac{e^{\frac{j\pi n}{4}} - e^{-\frac{j\pi n}{4}}}{2j} \right) z^{-n}$$

$$= \frac{1}{2j} \sum_{n=0}^{\infty} \left(\frac{1}{3} e^{\frac{j\pi}{4}} z^{-1}\right)^n - \frac{1}{2j} \sum_{n=0}^{\infty} \left(\frac{1}{3} e^{-\frac{j\pi}{4}} z^{-1}\right)^n$$

$$x[z] = \frac{1}{2j} \left(\frac{1}{1 - \frac{1}{3} e^{\frac{j\pi}{4}} z^{-1}} \right) - \frac{1}{2j} \left(\frac{1}{1 - \frac{1}{3} e^{-\frac{j\pi}{4}} z^{-1}} \right)$$

ROC₁:

$$\left| \frac{1}{3} e^{\frac{j\pi}{4}} z^{-1} \right| < 1$$

$$e^{\pm\theta} = \cos \theta \pm j \sin \theta$$

$$e^{\frac{j\pi}{4}} = \cos \frac{\pi}{4} + j \sin \frac{\pi}{4}$$

$$= \frac{\sqrt{2}}{2} + j \frac{\sqrt{2}}{2}$$

$$\left| e^{\frac{j\pi}{4}} \right| = \sqrt{\left(\frac{\sqrt{2}}{2}\right)^2 + \left(\frac{\sqrt{2}}{2}\right)^2} = 1$$

$$\left| \frac{1}{3z} \right| < 1$$

$$|z| > \frac{1}{3}$$

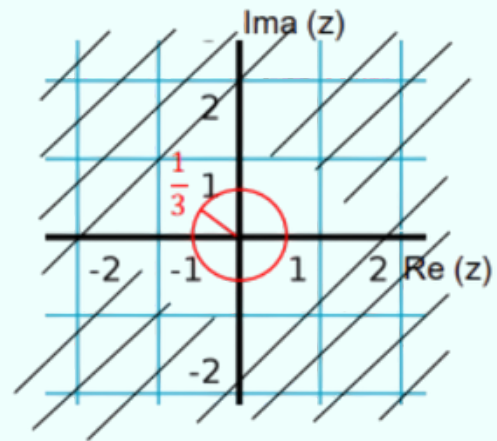
ROC₂:

$$\left| \frac{1}{3} e^{-\frac{j\pi}{4}} z^{-1} \right| < 1$$

$$e^{\pm\theta} = \cos \theta \pm j \sin \theta$$

$$\left| \frac{1}{3z} \right| < 1$$

$$|z| > \frac{1}{3}$$



INVERSE Z-TRANSFORM



- The inverse Z-transform is defined as the process of finding the time domain signal $x[n]$ from its Z-transform $X(z)$.

When using the **z-transform**

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$$

it is often useful to be able to find $x[n]$ given $X(z)$. There are at least 4 different methods to do this:

1. Inspection
2. Partial-Fraction Expansion
3. Power Series Expansion
4. Contour Integration



1

$$X(z) = \frac{\frac{1}{4}z^{-1}}{(1 - \frac{1}{2}z^{-1})(1 - \frac{1}{4}z^{-1})}$$

$$X(z) = \frac{A}{1 - \frac{1}{2}z^{-1}} + \frac{B}{1 - \frac{1}{4}z^{-1}}$$

$$A = \frac{\frac{1}{4}z^{-1}}{1 - \frac{1}{4}z^{-1}} \Big|_{z^{-1}=2} = \frac{\frac{1}{4}(2)}{1 - \frac{1}{4}(2)} = \frac{0.5}{1 - 0.5} = \frac{0.5}{0.5} = 1$$

$$B = \frac{\frac{1}{4}z^{-1}}{1 - \frac{1}{2}z^{-1}} \Big|_{z^{-1}=4} = \frac{\frac{1}{4}(4)}{1 - \frac{1}{2}(4)} = \frac{1}{1 - 2} = \frac{1}{-1} = -1$$

$$X(z) = \frac{1}{1 - \frac{1}{2}z^{-1}} - \frac{1}{1 - \frac{1}{4}z^{-1}}$$

$$x[n] = \left(\frac{1}{2}\right)^n u[n] - \left(\frac{1}{4}\right)^n u[n]$$

$$x[n] = -\left(\frac{1}{2}\right)^n u[-n-1] - \left[-\left(\frac{1}{4}\right)^n u[-n-1]\right]$$

$$x[n] = -\left(\frac{1}{2}\right)^n u[-n-1] + \left(\frac{1}{4}\right)^n u[-n-1]$$

$$x[n] = \underbrace{-\left(\frac{1}{2}\right)^n u[-n-1]}_{\text{Pole } 1/2 \text{ (Anti-causal)}} - \underbrace{\left(\frac{1}{4}\right)^n u[n]}_{\text{Pole } 1/4 \text{ (Causal)}}$$

2

$$X(z) = \frac{3 + 2z^{-1} + z^{-2}}{1 - 3z^{-1} + 2z^{-2}}$$

$$1 - 3z^{-1} + 2z^{-2} = (1 - z^{-1})(1 - 2z^{-1})$$

$$X(z) = C + \frac{A}{1 - z^{-1}} + \frac{B}{1 - 2z^{-1}}$$

$$C = \frac{\text{Coeff of } z^{-2} \text{ in Num}}{\text{Coeff of } z^{-2} \text{ in Den}} = \frac{1}{2} = 0.5$$

$$A = \frac{3 + 2z^{-1} + z^{-2}}{1 - 2z^{-1}} \Big|_{z^{-1}=1} = \frac{3 + 2(1) + 1}{1 - 2(1)} = \frac{6}{-1} = -6$$

$$B = \frac{3 + 2z^{-1} + z^{-2}}{1 - z^{-1}} \Big|_{z^{-1}=0.5} = \frac{3 + 2(0.5) + (0.5)^2}{1 - 0.5} = \frac{3 + 1 + 0.25}{0.5} = \frac{4.25}{0.5} = 8.5$$

$$X(z) = 0.5 - \frac{6}{1 - z^{-1}} + \frac{8.5}{1 - 2z^{-1}}$$

$$x[n] = 0.5\delta[n] - 6(1)^n u[n] + 8.5(2)^n u[n]$$

3

$$X(z) = \frac{1 + 3z^{-1}}{1 + 3z^{-1} + 2z^{-2}}$$

$$X(z) = \frac{(1 + 3z^{-1}) \cdot z^2}{(1 + 3z^{-1} + 2z^{-2}) \cdot z^2}$$

$$X(z) = \frac{z^2 + 3z}{z^2 + 3z + 2}$$

$$X(z) = \frac{z(z + 3)}{z^2 + 3z + 2}$$

$$\frac{X(z)}{z} = \frac{z + 3}{z^2 + 3z + 2}$$

$$z^2 + 3z + 2 = (z + 1)(z + 2)$$

$$\frac{X(z)}{z} = \frac{z + 3}{(z + 1)(z + 2)} = \frac{A}{z + 1} + \frac{B}{z + 2}$$

$$A = \left. \frac{z + 3}{z + 2} \right|_{z=-1} = \frac{-1 + 3}{-1 + 2} = \frac{2}{1} = 2$$

$$B = \left. \frac{z + 3}{z + 1} \right|_{z=-2} = \frac{-2 + 3}{-2 + 1} = \frac{1}{-1} = -1$$

$$\frac{X(z)}{z} = \frac{2}{z + 1} + \frac{-1}{z + 2}$$

$$x[n] = 2(-1)^n u[n] - (-2)^n u[n]$$

5

$$X(z) = \frac{1 + 2z^{-1} + z^{-2}}{1 - \frac{3}{2}z^{-1} + \frac{1}{2}z^{-2}}$$

$$\text{Remainder} = (1 + 2z^{-1} + z^{-2}) - 2 \left(1 - \frac{3}{2}z^{-1} + \frac{1}{2}z^{-2} \right)$$

$$\text{Remainder} = (1 + 2z^{-1} + z^{-2}) - (2 - 3z^{-1} + z^{-2})$$

$$\text{Remainder} = (1 - 2) + (2 - (-3))z^{-1} + (1 - 1)z^{-2}$$

$$\text{Remainder} = -1 + 5z^{-1}$$

$$X(z) = 2 + \frac{-1 + 5z^{-1}}{1 - \frac{3}{2}z^{-1} + \frac{1}{2}z^{-2}}$$

$$8u[n] - 9 \left(\frac{1}{2} \right)^n u[n]$$

$$x[n] = 2\delta[n] + 8u[n] - 9 \left(\frac{1}{2} \right)^n u[n]$$

4

$$X(z) = \frac{-1 + 5z^{-1}}{1 - \frac{3}{2}z^{-1} + \frac{1}{2}z^{-2}}$$

$$X(z) = \frac{(-1 + 5z^{-1}) \cdot z^2}{(1 - \frac{3}{2}z^{-1} + \frac{1}{2}z^{-2}) \cdot z^2}$$

$$X(z) = \frac{-z^2 + 5z}{z^2 - \frac{3}{2}z + \frac{1}{2}}$$

$$X(z) = \frac{z(-z + 5)}{z^2 - \frac{3}{2}z + \frac{1}{2}}$$

$$z^2 - \frac{3}{2}z + \frac{1}{2} = (z - 1)(z - \frac{1}{2})$$

$$\frac{X(z)}{z} = \frac{-z + 5}{(z - 1)(z - \frac{1}{2})} = \frac{A}{z - 1} + \frac{B}{z - \frac{1}{2}}$$

$$A = \left. \frac{-z + 5}{z - \frac{1}{2}} \right|_{z=1} = \frac{-1 + 5}{1 - 0.5} = \frac{4}{0.5} = 8$$

$$B = \left. \frac{-z + 5}{z - 1} \right|_{z=0.5} = \frac{-0.5 + 5}{0.5 - 1} = \frac{4.5}{-0.5} = -9$$

$$\frac{X(z)}{z} = \frac{8}{z - 1} + \frac{-9}{z - \frac{1}{2}}$$

$$x[n] = 8u[n] - 9 \left(\frac{1}{2} \right)^n u[n]$$

INVERSE Z-TRANSFORM USING POWER SERIES FUNCTION

- If ROC is greater than any number it is called casual. The denominator must be arranged from highest power to lowest power.
- If ROC is less than any number it is called non-casual. The denominator must be arranged from lowest to highest power.

Power Series?



Example



1 Find the IZT of $X(z) = \frac{1}{1-1.5z^{-1}+0.5z^{-2}}$ ROC: $|z| > 1$

$$\begin{array}{r} 1 + 1.5z^{-1} + 1.75z^{-2} + 1.875z^{-3} \\ 1 - 1.5z^{-1} + 0.5z^{-2} \overline{) 1} \\ 1 - 1.5z^{-1} + 0.5z^{-2} \\ \hline 1.5z^{-1} - 2.25z^{-2} + 0.75z^{-3} \\ 1.5z^{-1} - 2.25z^{-2} + 0.75z^{-3} \\ \hline 1.75z^{-2} - 0.75z^{-3} \\ 1.75z^{-2} - 2.625z^{-3} + 0.875z^{-4} \\ \hline 1.875z^{-3} - 0.875z^{-4} \end{array}$$

$$x[z] = 1 + 1.5z^{-1} + 1.75z^{-2} + 1.875z^{-3} + \dots$$

$$x[n] = \delta[n] + 1.5\delta[n-1] + 1.75\delta[n-2] + 1.875\delta[n-3] + \dots$$

$$x[n] = \{1, 1.5, 1.75, 1.875, \dots\}$$

2 Find the IZT of $X(z) = \frac{1}{1-1.5z^{-1}+0.5z^{-2}}$ ROC: $|z| < 0.5$

$$\begin{array}{r} 2z^2 + 6z^3 + 14z^4 + 30z^5 \\ 0.5z^{-2} - 1.5z^{-1} + 1 \overline{) 1} \\ 1 - 3z + 2z^2 \\ \hline 3z - 2z^2 \\ 3z - 9z^2 + 6z^3 \\ \hline 7z^2 - 6z^3 \\ 7z^2 - 21z^3 + 14z^4 \\ \hline 15z^3 - 14z^4 \end{array}$$

$$x[z] = 2z^2 + 6z^3 + 14z^4 + 30z^5 + \dots$$

$$x[n] = 2\delta[n+2] + 6\delta[n+3] + 14\delta[n+4] + 30\delta[n+5] + \dots$$

$$x[n] = \{\dots, 30, 14, 6, 2, 0, 0\}$$

3 Find the IZT of $X(z) = \frac{z}{2z^2-3z+1}$ ROC: $|z| > \frac{1}{2}$

$$\begin{array}{r} \frac{1}{2}z^{-1} + \frac{3}{4}z^{-2} + \frac{7}{8}z^{-3} \\ 2z^2 - 3z + 1 \overline{) z} \\ z - \frac{3}{2} + \frac{1}{2}z^{-1} \\ \hline \frac{3}{2} - \frac{1}{2}z^{-1} \\ \frac{3}{2} - \frac{9}{4}z^{-1} + \frac{3}{4}z^{-2} \\ \hline \frac{7}{4}z^{-1} - \frac{3}{4}z^{-2} \\ \frac{7}{4}z^{-1} - \frac{21}{8}z^{-2} + \frac{7}{8}z^{-3} \\ \hline \frac{15}{8}z^{-2} - \frac{7}{8}z^{-3} \end{array}$$

$$x[z] = \frac{1}{2}z^{-1} + \frac{3}{4}z^{-2} + \frac{7}{8}z^{-3} + \dots$$

$$x[n] = \frac{1}{2}\delta[n-1] + \frac{3}{4}\delta[n-2] + \frac{7}{8}\delta[n-3] + \dots$$

$$x[n] = \left\{0, \frac{1}{2}, \frac{3}{4}, \frac{7}{8}, \dots\right\}$$

4 $X(z) = \frac{z}{2z^2-3z+1}$ ROC: $|z| > \frac{1}{2}$

$$\begin{array}{r} z + 3z^2 + 7z^3 \\ 1 - 3z + 2z^2 \overline{) z} \\ z - 3z^2 + 2z^3 \\ \hline 3z^2 - 2z^3 \\ 3z^2 - 9z^3 + 6z^4 \\ \hline 7z^3 - 6z^4 \\ 7z^3 - 21z^4 + 14z^5 \\ \hline 15z^4 - 14z^5 \end{array}$$

$$x[z] = z + 3z^2 + 7z^3 + \dots$$

$$x[n] = \delta[n+1] + 3\delta[n+2] + 7\delta[n+3] + \dots$$

$$x[n] = \{\dots, 7, 3, 1, 0\}$$

FILTERS

Filters!



- series of components designed to reduce the amplitude or attenuate of an unwanted frequency in respect to some cutoff frequency

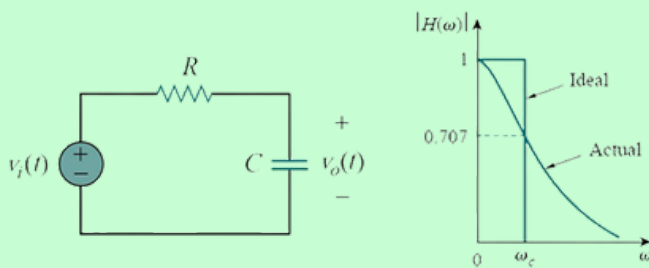
- passes lower frequencies and stops high frequencies

Low Pass



Lowpass Filters

A **lowpass filter** is designed to pass only frequencies from dc up to the cutoff frequency ω_c



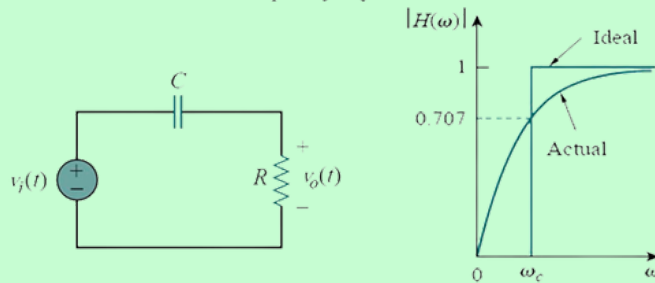
High Pass



- pass high frequencies and rejects low frequencies

Highpass Filter

A **highpass filter** is designed to pass all frequencies above its cutoff frequency ω_c



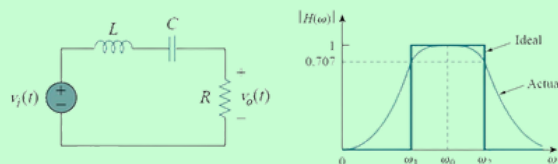
Bandpass



- passes frequencies within a frequency band and blocks or attenuates frequencies outside the band

Bandpass Filter

A **bandpass filter** is designed to pass all frequencies within a band of frequencies, $\omega_1 < \omega_0 < \omega_2$



Bandstop



- passes frequencies outside a frequency band and blocks or attenuates frequencies within the band

Bandstop Filter

A **bandstop filter** is designed to stop or eliminate all frequencies within a band of frequencies, $\omega_1 < \omega_0 < \omega_2$

